# Module III — Semiconductor properties

Derivations, step by step

## Conductivity of a semiconductor: $\sigma = e(n\mu_n + p\mu_p)$

4 marks

$v_d$: drift velocity. $\mu$: mobility, the drift velocity per unit field. $n$, $p$ : electron and hole concentrations; $\mu_n$, $\mu_p$ their mobilities. $\mathcal{E}$: applied field.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $v_d$ | drift velocity, the average velocity gained from the field: Drude derivation |
| $\mu$ | mobility, drift velocity per unit field: drift velocity and mobility |
| $n$, $p$ | electrons in the conduction band and holes in the valence band: intrinsic semiconductors |
| $J = \sigma\mathcal{E}$ | Ohm's law in local form: the conductivity $\sigma$ is defined by it |
| $e$ | magnitude of the electron charge; holes carry $+e$: holes |

START

$$v_d = \mu\mathcal{E}$$

RESULT

$$\sigma_i = en_i(\mu_n + \mu_p)$$

### DRIFT

$$v_d = \mu\mathcal{E}$$

Carriers are accelerated by the field but keep colliding with the lattice (phonons, impurities). On average they gain only a small extra velocity, proportional to the field: $v_d = e\tau\mathcal{E}/m$ with the relaxation time $\tau$, as in the Drude derivation. The constant $e\tau/m$ is the mobility $\mu$.

### ELECTRON CURRENT

$$\text{charge through area } A \text{ in } dt = n\,e\,(A\,v_{dn}\,dt)$$

*Every carrier within $v_{dn}\,dt$ of the area gets through in time $dt$*

They fill a box of base $A$ and length $v_{dn}\,dt$, which holds $nAv_{dn}\,dt$ carriers of charge $e$ each.

$$J_n = nev_{dn}$$

*Current per unit area: divide the charge by $A\,dt$*

$n$ electrons per unit volume, each carrying charge $e$ at the drift velocity.

$$J_n = ne\mu_n\mathcal{E}$$

*substitute: $v_{dn} = \mu_n\mathcal{E}$*

$$\sigma_n = ne\mu_n$$

*Compare with Ohm's law $J = \sigma\mathcal{E}$*

### HOLES ADD TO IT

$$\sigma = \sigma_n + \sigma_p = e(n\mu_n + p\mu_p)$$

*add: $\sigma_p = pe\mu_p$*

Holes (what a hole is) drift the other way, but their charge is $+e$, so their current is in the same direction as the electrons' current. The two conductivities add.

**Check.** Electrons and holes drift in opposite directions. Do their currents add or cancel?

- ✓ **add: opposite charge and opposite motion give current in the same direction**
- cancel exactly
- only the faster one counts
- they cancel in an intrinsic sample

Current is charge × velocity; both signs flip for holes, so the product has the same sign.

### INTRINSIC SAMPLE

$$\sigma_i = en_i(\mu_n + \mu_p)$$

*substitute: $n = p = n_i$*

$\mu_n > \mu_p$ because a hole moves by electrons hopping between bonds and has the larger effective mass. Si at 300 K: $\mu_n = 0.135$, $\mu_p = 0.048\ \mathrm{m^2V^{-1}s^{-1}}$, $n_i = 1.5 \times 10^{16}\ \mathrm{m^{-3}}$ give $\sigma_i = 4.4 \times 10^{-4}\ \mathrm{S\,m^{-1}}$.

## $\ln\sigma$ against $1/T$ measures the band gap

4 marks

Mobilities taken as independent of temperature. $n_i$ from the law of mass action.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $\sigma_i = en_i(\mu_n + \mu_p)$ | intrinsic conductivity: conductivity derivation |
| $n_i = (N_cN_v)^{1/2}e^{-E_g/2kT}$ | from the law of mass action |
| $\mu_n$, $\mu_p$ | mobilities, taken as independent of temperature here |
| $N_cN_v \propto T^3$ | shown in the step below |

START

$$\sigma = en_i(\mu_n + \mu_p)$$

RESULT

$$\ln\sigma = -\frac{E_g}{2k}\cdot\frac{1}{T} + \frac{3}{2}\ln T + \text{const}$$

### START

$$\sigma = en_i(\mu_n + \mu_p)$$

The intrinsic conductivity.

$$\sigma = e(\mu_n + \mu_p)(N_cN_v)^{1/2}e^{-E_g/2kT}$$

*substitute: $n_i$*

$$(N_c N_v)^{1/2} = 2\left(\frac{2\pi k}{h^2}\right)^{3/2} (m_n^* m_p^*)^{3/4}\, T^{3/2}$$

*Put in $N_c$ and $N_v$ (electron, hole derivations): each carries $T^{3/2}$*

Only $T$ changes with temperature; the masses are constants.

$$\sigma = C T^{3/2} e^{-E_g/2kT}$$

$N_c N_v \propto T^3$*: collect the constants into* $C$

**TAKE LOGS**

$$\ln \sigma = \ln C + \frac{3}{2}\ln T - \frac{E_g}{2kT}$$

*ln*

$$\ln \sigma = -\frac{E_g}{2k}\cdot\frac{1}{T} + \frac{3}{2}\ln T + \text{const}$$

*rearrange*

The $1/T$ term dominates: $\ln\sigma$ against $1/T$ is nearly a straight line of slope $-E_g/2k$. This is how a band gap is measured.

**Check.** The slope of $\ln\sigma$ against $1/T$ is $-5800$ K. What is $E_g$?

- ✓ $2k \times 5800\ \text{K} \approx 1.0$ eV
- $k \times 5800\ \text{K} \approx 0.5$ eV
- $5800\ \text{K}/2k$
- $0.26$ eV

Slope $= -E_g/2k$, so $E_g = 2k \times 5800 = 2 \times 8.62\times10^{-5} \times 5800 \approx 1.0$ eV.

## Electron concentration in the conduction band

8 marks

$D(E)$: density of states per unit volume. $f(E)$: Fermi function. $m_n^*$: effective mass of electrons in the conduction band. $E_c$: bottom of the conduction band. Kinetic energy of an electron at $E$ is $E - E_c$.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $D(E)$ | states per unit energy and volume, from the density-of-states derivation (the version with $h$ is its last step) |
| $f(E)$ | Fermi function, probability that a state is occupied: the Fermi function |
| $E_c$, $E_v$, $E_g$ | band edges and gap: metals, semiconductors, insulators |
| $m_n^*$ | effective mass of an electron in the conduction band: effective-mass derivation |
| $E_F$ | Fermi level, lying inside the gap: the recipe: states times occupation |
| $\int_0^\infty x^{1/2}e^{-x}dx$ | the integral is evaluated in the steps below |

START

$$dn = D(E)f(E)\,dE$$

RESULT

$$n = N_c\, e^{-(E_c - E_F)/kT}$$

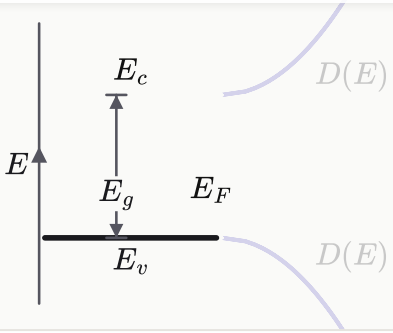
$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

**COUNT OCCUPIED STATES**

$$dn = D(E)f(E)\,dE$$

Electrons between $E$ and $E + dE$ = (states there) × (chance each is occupied).

Electrons fill the conduction band from $E_c$ up

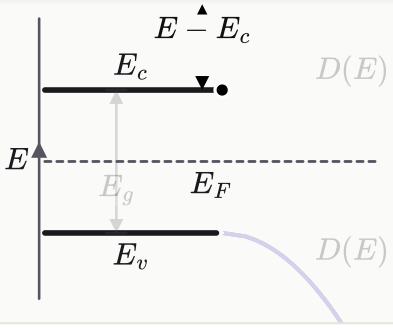



$$D(E) = \frac{4\pi}{h^3}(2m_n^*)^{3/2}(E - E_c)^{1/2}$$

*Free-electron $D(E)$, with $m \to m_n^*$ and $E \to E - E_c$*

The free-electron result $\frac{4\pi}{h^3}(2m)^{3/2}E^{1/2}$ per unit volume (derived in Module II) counts kinetic energy from zero. In the conduction band the electron behaves as a free particle of mass $m_n^*$ (effective mass) with kinetic energy counted from $E_c$.

$D(E) \propto \sqrt{E - E_c}$



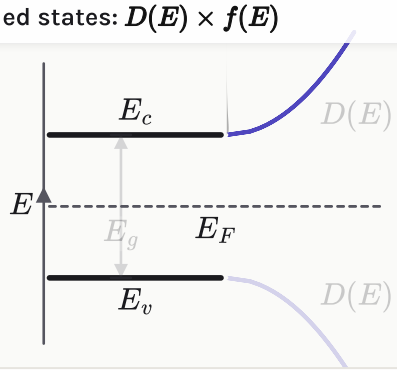
### BOLTZMANN TAIL

$$f(E) = \frac{1}{e^{(E-E_F)/kT}+1} \approx e^{-(E-E_F)/kT}$$

*For $E - E_F \gg kT$, drop the 1*

$E_F$ lies in the gap, several $kT$ below $E_c$ ($kT = 0.026$ eV at 300 K), so the exponential is huge for every conduction-band state and the 1 is negligible. This is the classical Boltzmann form of the Fermi function.

Occupied states: $D(E) \times f(E)$




**Check.** Why can we drop the $+1$ in the Fermi function here?

- ✓ $E - E_F \gg kT$ for every state in the conduction band
- $E_F$ is above $E_c$
- $kT$ is large
- the band is full

With $E - E_F$ several $kT$, $e^{(E-E_F)/kT}$ is very much greater than 1.

### INTEGRATE OVER THE BAND

$$n = A\int_{E_c}^{\infty} (E-E_c)^{1/2} e^{-(E-E_F)/kT}\,dE$$

*Add up from $E_c$ to $\infty$, writing $A = \frac{4\pi}{h^3}(2m_n^*)^{3/2}$*

$$n = Ae^{-(E_c-E_F)/kT}\int_{E_c}^{\infty} (E-E_c)^{1/2} e^{-(E-E_c)/kT}\,dE$$

*Split $E - E_F = (E - E_c) + (E_c - E_F)$ and take the constant out*

### SUBSTITUTE $x = (E - E_c)/kT$

$$n = A(kT)^{3/2} e^{-(E_c-E_F)/kT}\int_0^{\infty} x^{1/2} e^{-x}\,dx$$

*substitute: $E - E_c = kTx,\ dE = kT\,dx$*

$(E-E_c)^{1/2}dE = (kT)^{3/2}x^{1/2}dx$; the limits become 0 and $\infty$.

### THE INTEGRAL OVER $x$

$$\int_0^{\infty} x^{1/2}e^{-x}dx = \frac{1}{2}\int_0^{\infty} x^{-1/2}e^{-x}dx$$

*By parts: $u = x^{1/2}$, $dv = e^{-x}dx$*

The boundary term $\left[-x^{1/2}e^{-x}\right]_0^{\infty}$ is zero and what is left is $\int_0^{\infty} \frac{1}{2}x^{-1/2}e^{-x}dx$.

$$\int_0^{\infty} x^{-1/2}e^{-x}dx = 2\int_0^{\infty} e^{-y^2}dy$$

*substitute: $x = y^2,\ dx = 2y\,dy$*

$$\int_0^{\infty} e^{-y^2}dy = \frac{\sqrt{\pi}}{2}$$

*The Gaussian integral*

Call it $I$ and square it: $I^2 = \int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)}dx\,dy$. In polar coordinates over the quarter plane, $I^2 = \int_0^{\pi/2} d\phi \int_0^{\infty} e^{-r^2} r\,dr = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}$.

$$\int_0^{\infty} x^{1/2}e^{-x}dx = \frac{1}{2}\cdot 2\cdot\frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{2}$$

*combine*

$$n = A(kT)^{3/2}e^{-(E_c-E_F)/kT}\cdot\frac{\sqrt{\pi}}{2}$$

*evaluate: $\int_0^{\infty} x^{1/2}e^{-x}dx = \sqrt{\pi}/2$*

Put the value just derived into the integral.

### TIDY

$$n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} e^{-(E_c-E_F)/kT}$$

*simplify*

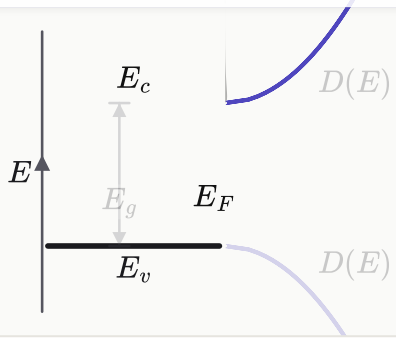
$A(kT)^{3/2}\frac{\sqrt{\pi}}{2} = \frac{4\pi}{h^3}\cdot\frac{\sqrt{\pi}}{2}(2m_n^*kT)^{3/2} = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$.

$$n = N_c\, e^{-(E_c-E_F)/kT}$$

*define: $N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$*

$N_c$ is the effective density of states at the band edge: it is as if all the states were squeezed onto $E_c$, each occupied with probability $e^{-(E_c-E_F)/kT}$. For Si, $N_c \approx 2.8\times10^{25}(T/300)^{3/2}\ \mathrm{m}^{-3}$. The result holds for doped samples too.

$n = N_c e^{-(E_c-E_F)/kT}$




## Hole concentration in the valence band

5 marks

$1 - f(E)$: probability that a state is empty, i.e. holds a hole. $m_p^*$: hole effective mass. $E_v$: top of the valence band; a hole at $E$ has kinetic energy $E_v - E$.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $1 - f(E)$ | probability that a state is empty, that is, holds a hole: holes |
| $m_p^*$ | hole effective mass: holes |
| $E_v$ | top of the valence band; a hole at $E$ has kinetic energy $E_v - E$ |
| $D(E)$, $\sqrt{\pi}/2$ | same density of states and same integral as in the electron derivation |

START

$$dp = D(E)\,[1 - f(E)]\,dE$$

RESULT

$$p = N_v\, e^{-(E_F - E_v)/kT}$$

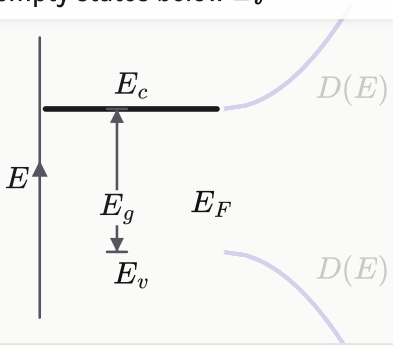
$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

## COUNT EMPTY STATES

$$dp = D(E)\,[1 - f(E)]\,dE$$

A hole is a state the electrons have left empty.

Holes: empty states below $E_v$




## PROBABILITY OF AN EMPTY STATE

$$1 - f(E) = 1 - \frac{1}{e^{(E-E_F)/kT} + 1} = \frac{e^{(E-E_F)/kT}}{e^{(E-E_F)/kT} + 1}$$

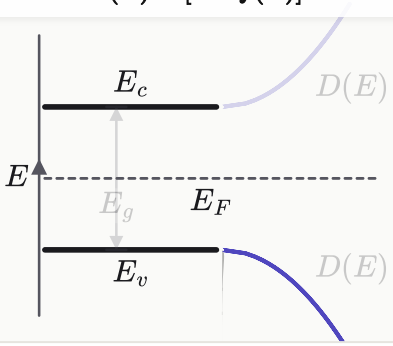
*Put over a common denominator*

$$1 - f(E) = \frac{1}{e^{(E_F-E)/kT} + 1} \approx e^{-(E_F-E)/kT}$$

*For $E_F - E \gg kT$*

Divide top and bottom by $e^{(E-E_F)/kT}$ to get $1/(e^{(E_F-E)/kT} + 1)$. Holes become exponentially rarer as you go deeper into the valence band.

Empty states: $D(E) \times [1 - f(E)]$




**Check.** In the valence band $E < E_F$. Which form does $1 - f(E)$ take?

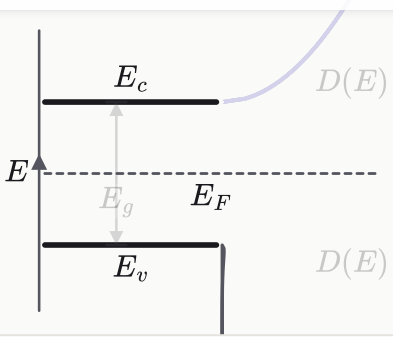
- ✓ $e^{-(E_F-E)/kT}$
- $e^{-(E-E_F)/kT}$
- $1$
- $e^{-E/kT}$

$1 - \frac{1}{e^x+1} = \frac{1}{e^{-x}+1}$ with $x = (E - E_F)/kT < 0$; for $|x| \gg 1$ this is $e^x = e^{-(E_F-E)/kT}$.

$$D(E) = \frac{4\pi}{h^3}(2m_p^*)^{3/2}(E_v - E)^{1/2}$$

*Same $D(E)$, with $m_p^*$ and kinetic energy $E_v - E$*

$D(E) \propto \sqrt{E_v - E}$




## INTEGRATE DOWN THE BAND

$$p = A' \int_{-\infty}^{E_v} (E_v - E)^{1/2} e^{-(E_F-E)/kT}\, dE$$

*Add up from $-\infty$ to $E_v$, with $A' = \frac{4\pi}{h^3}(2m_p^*)^{3/2}$*

$$p = A'(kT)^{3/2} e^{-(E_F-E_v)/kT} \cdot \frac{\sqrt{\pi}}{2}$$

*substitute: $E_v - E = kTx$*

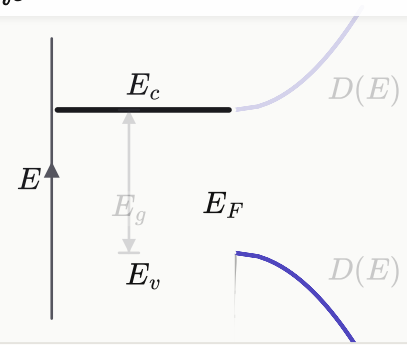
The same integral as for electrons, derived in the electron derivation.

$$p = N_v\, e^{-(E_F - E_v)/kT}$$

*define:* $N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$

The mirror image of the electron result, with $m_p^*$ and the distance of $E_F$ above $E_v$.

$p = N_v e^{-(E_F - E_v)/kT}$




## Fermi level of an intrinsic semiconductor

5 marks

Intrinsic: $n = p = n_i$. $N_c$, $N_v$ as in the two previous derivations.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $n = p = n_i$ | in a pure sample every electron in the conduction band left a hole: intrinsic semiconductors |
| $N_c$, $N_v$ | effective densities of states from the electron and the hole derivations |
| $E_F$ | Fermi level: where it sits in a pure sample |
| $\ln x^a = a \ln x$ | the logarithm rule used in the step with the ask |

START

$$N_c e^{-(E_c - E_F)/kT} = N_v e^{-(E_F - E_v)/kT}$$

RESULT

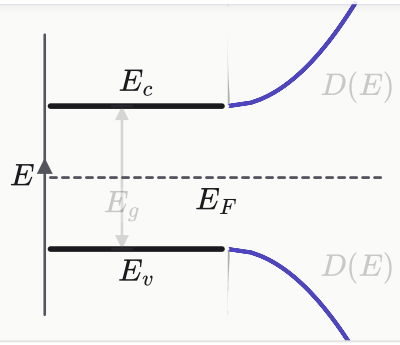
$$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4} \ln \frac{m_p^*}{m_n^*}$$

**N = P**

$$N_c e^{-(E_c - E_F)/kT} = N_v e^{-(E_F - E_v)/kT}$$

In a pure sample every conduction electron left a hole behind.

$n = p$




$$e^{(2E_F - E_c - E_v)/kT} = \frac{N_v}{N_c}$$

*Collect the exponentials on one side*

$$\frac{2E_F - E_c - E_v}{kT} = \ln \frac{N_v}{N_c}$$

*ln*

$$E_F = \frac{E_c + E_v}{2} + \frac{kT}{2} \ln \frac{N_v}{N_c}$$

*rearrange*

**PUT IN $N_c$ AND $N_v$**

$$\frac{N_v}{N_c} = \left(\frac{m_p^*}{m_n^*}\right)^{3/2}$$

*Everything in $N_c$, $N_v$ cancels except the masses*

**Check.** What is $\ln\left[(m_p^*/m_n^*)^{3/2}\right]$?

- ✓ $\frac{3}{2}\ln(m_p^*/m_n^*)$
- $\ln(m_p^*/m_n^*)^{2/3}$
- $(\ln m_p^* / \ln m_n^*)^{3/2}$
- $\frac{2}{3}\ln(m_p^*/m_n^*)$

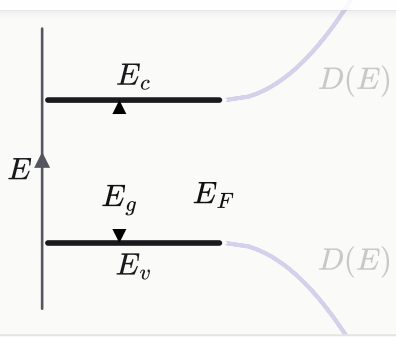
$\ln x^a = a \ln x$.

$$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4} \ln \frac{m_p^*}{m_n^*}$$

*substitute*

At 0 K, or whenever $m_p^* = m_n^*$, $E_F$ is exactly at mid-gap. Usually $m_p^* > m_n^*$, so $E_F$ rises slightly with $T$; for Si at 300 K by only about 0.01 eV.

$E_F$ at the middle of the gap




## Law of mass action and the intrinsic concentration $n_i$

5 marks

$E_g = E_c - E_v$: band gap. $n$, $p$ from the two concentration derivations.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $n$, $p$ | from the electron and the hole derivations |
| $E_g = E_c - E_v$ | band gap: metals, semiconductors, insulators |
| $n_i$ | intrinsic carrier concentration: intrinsic semiconductors |
| **mass action** | $np$ fixed at a given temperature: the law and $n_i$ |

START

$$np = N_c N_v e^{-(E_c - E_F)/kT} e^{-(E_F - E_v)/kT}$$

RESULT

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} (m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

## MULTIPLY N AND P

$$np = N_c N_v e^{-(E_c - E_F)/kT} e^{-(E_F - E_v)/kT}$$

Multiply the two results.

$$np = N_c N_v e^{-(E_c - E_v)/kT}$$

*combine*

$E_F$ cancels.

**Check.** What happened to $E_F$?

- ✓ it cancelled: $-(E_c - E_F) - (E_F - E_v) = -(E_c - E_v)$
- it was set to zero
- it equals $E_c$
- it doubles

Add the two exponents: the $E_F$ terms are $+E_F$ and $-E_F$.

$$np = N_c N_v e^{-E_g/kT} = n_i^2$$

*substitute:* $E_c - E_v = E_g$

**Law of mass action**: at a given temperature $np$ is fixed, whether the sample is doped or not. Doping that raises $n$ lowers $p$ in proportion.

## INTRINSIC CONCENTRATION

$$n_i = (N_c N_v)^{1/2} e^{-E_g/2kT}$$

*root*

Note the 2 in the exponent: $E_g/2kT$, not $E_g/kT$.

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} (m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

*Put in $N_c$ and $N_v$*

Pure Ge at 300 K: $n_i \approx 2.4 \times 10^{19}\ \mathrm{m}^{-3}$ out of $4.4 \times 10^{28}$ atoms per $\mathrm{m}^3$, about 5 broken bonds in $10^{10}$ atoms.

### n-type: Fermi level and electron concentration

8 marks

$N_d$: donors per unit volume at level $E_d$, just below $E_c$. $N_d^+$: ionised donors. $\Delta E = E_c - E_d$: ionisation energy of a donor. Moderate temperature: holes are negligible, and $E_F$ is a few $kT$ from both $E_c$ and $E_d$.

**BEFORE YOU START: WHAT THIS USES**

| | |
|---|---|
| $N_d$, $E_d$ | donor concentration and donor level just below $E_c$: extrinsic semiconductors |
| $N_d^+$ | ionised donors: donors that have lost their electron |
| $n = N_c e^{-(E_c - E_F)/kT}$ | electron concentration: the electron derivation |
| $p = n_i^2/n$ | law of mass action: mass-action derivation |
| $1 - f(E_d)$ | chance that the donor level is empty: how $1 - f$ becomes a Boltzmann factor |
| $\Delta E = E_c - E_d$ | ionisation energy of a donor |

START

$$n = N_d^+ + p \approx N_d^+$$

RESULT

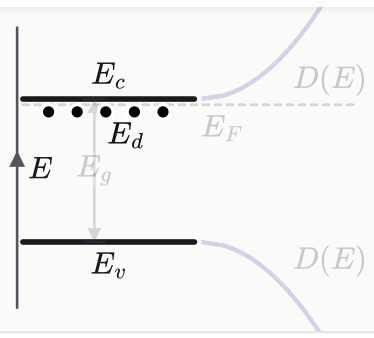
$$n = (N_d N_c)^{1/2} e^{-\Delta E/2kT}$$

## NEUTRALITY

$$n = N_d^+ + p \approx N_d^+$$

Every conduction electron came from an ionised donor or from a broken bond. With enough donors, $p = n_i^2/n$ is tiny (mass action), so drop it. This is charge neutrality (extrinsic semiconductors).

Donor levels $E_d$ just below $E_c$




## IONISED DONORS

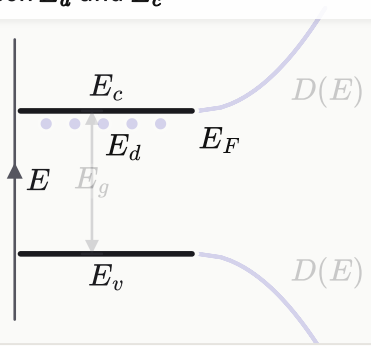
$$N_d^+ = N_d\,[1 - f(E_d)]$$

*A donor is ionised when its level is empty*

$$N_d^+ \approx N_d e^{-(E_F - E_d)/kT}$$

*Boltzmann form of* $1 - f$*, since* $E_F - E_d \gg kT$

**$E_F$ between $E_d$ and $E_c$**




**Check.** Is $E_F$ above or below $E_d$ here, and what does that give?

- ✓ above, so $1 - f(E_d) \approx e^{-(E_F - E_d)/kT}$
- below, so $1 - f \approx 1$
- equal, so $1 - f = \frac{1}{2}$
- it does not matter

At moderate $T$, $E_F$ lies between $E_d$ and $E_c$, a few $kT$ above $E_d$. Then $1 - f(E_d) = 1/(e^{(E_F - E_d)/kT} + 1) \approx e^{-(E_F - E_d)/kT}$.

## EQUATE WITH N

$$N_c e^{-(E_c - E_F)/kT} = N_d e^{-(E_F - E_d)/kT}$$

*substitute:* $n = N_c e^{-(E_c - E_F)/kT}$

$$e^{(2E_F - E_c - E_d)/kT} = \frac{N_d}{N_c}$$

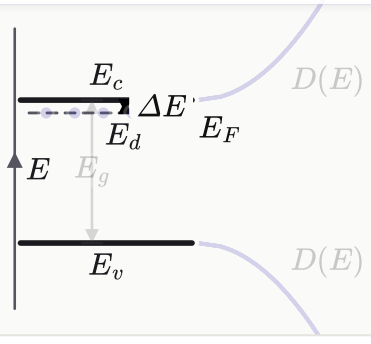
*rearrange*

$$E_F = \frac{E_c + E_d}{2} + \frac{kT}{2} \ln \frac{N_d}{N_c}$$

*Take* $\ln$ *and solve for* $E_F$

At moderate $T$, $E_F$ lies about midway between $E_d$ and $E_c$. As $T$ rises ($N_c \propto T^{3/2}$) it moves down and finally reaches mid-gap: the sample turns intrinsic. Not valid at $T = 0$ or very high $T$.

**$E_F \approx (E_c + E_d)/2$**




## ELECTRON CONCENTRATION

$$n = N_c \exp\left[-\frac{E_c - \frac{E_c + E_d}{2} - \frac{kT}{2}\ln\frac{N_d}{N_c}}{kT}\right]$$

*substitute:* $E_F$

$$n = N_c \left(\frac{N_d}{N_c}\right)^{1/2} e^{-(E_c - E_d)/2kT}$$

*simplify*

$$n = (N_d N_c)^{1/2} e^{-\Delta E/2kT}$$

*simplify*

$n \propto \sqrt{N_d}$ at moderate temperatures. Conductivity $\sigma = ne\mu_n \approx N_d e \mu_n$ once all donors are ionised. p-type, the same way: $E_F = \frac{E_v + E_a}{2} - \frac{kT}{2}\ln\frac{N_a}{N_v}$, $\sigma_p \approx N_a e \mu_p$.

**$\Delta E = E_c - E_d$**

